# === UNIT 6: TRANSLATION, GENETIC CODE, RNA PROCESSING & GENE REGULATION ===

## >> 6.1 The Genetic Code and Aminoacyl-tRNA Synthetases

Translation is the ribosomal decoding of messenger RNA (mRNA) nucleotide triplets into polypeptide chains of defined primary sequence.

* Genetic Code Characteristics: (1) Triplet: 3 consecutive ribonucleotides specify 1 amino acid (4^3 = 64 possible codons). (2) Degenerate (Redundant): 61 sense codons specify 20 standard amino acids; multiple synonymous codons encode the same amino acid (except Met [AUG] and Trp [UGG]). (3) Non-overlapping and Commaless: read sequentially without punctuation. (4) Universal: standard code is shared across all terrestrial organisms (with minor mitochondrial codon variations, e.g., UGA encoding Trp instead of Stop). (5) Initiation and Termination: AUG encodes N-formylmethionine (prokaryotes) or methionine (eukaryotes); UAA (Ochre), UAG (Amber), and UGA (Opal) act as stop codons.

* Crick's Wobble Hypothesis: Explains why cells require fewer than 61 distinct tRNA species (bacteria have ~30-40 tRNAs). The first two bases of the mRNA codon (5' end) form rigid, standard Watson-Crick pairs with the 3' positions of the tRNA anticodon. Base pairing at the third codon position (3' end) and first anticodon position (5' end) exhibits conformational flexibility ('wobble'):

- Anticodon 5' G pairs with Codon 3' C or U.
- Anticodon 5' U pairs with Codon 3' A or G.
- Anticodon 5' Inosine (I, deaminated adenine) pairs with Codon 3' A, U, or C.

* Aminoacyl-tRNA Synthetases (aaRS): The 'Second Genetic Code'. Two-step charging reaction: (1) Amino acid + ATP -> Aminoacyl-AMP + PPi (amino acid activation). (2) Aminoacyl-AMP + tRNA -> Aminoacyl-tRNA + AMP (esterification to 2'-OH or 3'-OH of terminal adenosine at the CCA-3' acceptor stem). Class I and Class II aaRS enzymes possess editing/proofreading pockets that hydrolyze incorrectly attached amino acids, maintaining mischarging rates below 1 in 10,000.

## >> 6.2 Ribosomal Translation Mechanism in Prokaryotes

* Ribosome Structure: Bacterial 70S ribosome composed of a 30S small subunit (16S rRNA + 21 proteins) and a 50S large subunit (23S rRNA, 5S rRNA + 31 proteins). Three functional tRNA binding sites: A (Aminoacyl) site, P (Peptidyl) site, and E (Exit) site.

* Translation Stages:

1. Initiation: IF3 binds 30S to prevent premature 50S association. IF1 blocks the A site. 16S rRNA aligns with the mRNA Shine-Dalgarno sequence (5'-AGGAGG-3') located 8 bp upstream of the AUG start codon. IF2-GTP recruits initiator formylmethionyl-tRNA (fMet-tRNAfMet) directly into the P site. 50S subunit docks, triggering IF2 GTP hydrolysis and release of all initiation factors.
2. Elongation: Ternary complex of EF-Tu-GTP-aminoacyl-tRNA delivers cognate

aminoacyl-tRNA to the A site. Correct codon-anticodon pairing triggers EF-Tu GTP hydrolysis and release. EF-Ts acts as a guanine nucleotide exchange factor (GEF) recycling EF-Tu-GDP to EF-Tu-GTP. Peptidyl Transferase Center (PTC) in the 23S rRNA (a ribozyme) catalyzes peptide bond formation: the alpha-amino group of the A-site amino acid attacks the ester carbonyl carbon of the P-site peptidyl-tRNA. EF-G-GTP binds and drives translocation: deacylated tRNA shifts from P to E site; newly formed peptidyl-tRNA shifts from A to P site; mRNA moves precisely 3 nucleotides.
3. Termination: When a stop codon (UAA, UAG, UGA) enters the A site, release factors RF1 (UAA/UAG) or RF2 (UAA/UGA) enter the A site. Their conserved GGQ tripeptide motif interacts with the PTC, inducing water molecule attack on the peptidyl-tRNA ester bond, releasing the completed polypeptide. RF3-GTP drives RF release, and Ribosome Recycling Factor (RRF) + EF-G disassemble the 70S ribosome into 30S and 50S subunits.

### >> 6.3 Eukaryotic Post-Transcriptional RNA Processing

Eukaryotic pre-mRNAs undergo co-transcriptional chemical modifications before nuclear export:
* 5’ Capping: Addition of 7-methylguanosine (m7G) linked via an unusual 5’-to-5’ triphosphate bridge. Performed by RNA triphosphatase, guanylyltransferase, and (guanine-N7)-methyltransferase. Protects mRNA from 5’ exonucleolytic decay, facilitates nuclear export via CBC, and serves as the binding platform for eIF4E during translation initiation.
* 3’ Polyadenylation: Cleavage and Polyadenylation Specificity Factor (CPSF) recognizes the hexanucleotide motif AAUAAA, while CstF recognizes a downstream G/U-rich element. Endonuclease cleaves RNA 10-30 nt downstream of AAUAAA. Poly(A) Polymerase (PAP) adds a tail of 200-250 adenosine residues without a template. Bound by PABP to enhance mRNA stability and stimulate translation through circularization.
* Splicing: Removal of non-coding introns and ligation of coding exons executed by the Spliceosome (macromolecular ribonucleoprotein complex of U1, U2, U4, U5, U6 snRNPs). Conserved splicing signals: 5’ splice site (GU), 3’ splice site (AG), and internal Branch Point adenosine (located 20-40 nt upstream of 3’ site within a polypyrimidine tract). Two-step transesterification: (1) 2’-OH of branch point A attacks 5’ splice site phosphate, generating a free 5’ exon and a lariat-intron intermediate. (2) Free 3’-OH of 5’ exon attacks 3’ splice site phosphate, ligating exons and releasing the excised lariat.

### >> 6.4 Bacterial Operon Regulation: Lac and Trp Operons

* The Lactose (Lac) Operon (Negative Inducible & Positive Catabolite Control):
- Structural Genes: lacZ (beta-galactosidase, cleaves lactose into glucose + galactose and isomerizes lactose to allolactose), lacY (lactose permease), lacA (thiogalactoside transacetylase).
- Repressor Control: lacI encodes homotetrameric Lac repressor which binds the lac operator (lacO), looping DNA and sterically preventing RNA Pol promoter clearance.

When lactose is present, allolactose binds the repressor, inducing an allosteric conformational shift that causes it to dissociate from lacO (derepression).
- Catabolite Repression (CAP-cAMP): Glucose is the preferred carbon source. High glucose inhibits adenylate cyclase, maintaining low cAMP. When glucose is depleted, adenylate cyclase synthesizes cAMP. cAMP binds the Catabolite Activator Protein (CAP / CRP). The CAP-cAMP dimer binds a site upstream of the promoter (-61.5), bending DNA by ~90 degrees and making direct protein-protein contacts with the alpha-CTD of RNA Polymerase, stimulating transcription >50-fold. Maximum expression requires lactose present AND glucose absent.
* The Tryptophan (Trp) Operon (Negative Repressible & Attenuation Control):
- Structural Genes: trpE, trpD, trpC, trpB, trpA (biosynthetic enzymes for tryptophan).
- Repressor: TrpR aporepressor binds corepressor tryptophan. Only the TrpR-tryptophan complex binds the trp operator, repressing transcription 70-fold when tryptophan is abundant.
- Attenuation Mechanism: Fine-tunes expression an additional 10-fold by sensing charged tRNATrp levels during translation of a 14-amino-acid leader peptide (trpL) containing two consecutive Trp codons (UGG-UGG). The leader mRNA can form alternative secondary hairpins: Region 1:2 hairpin, Region 2:3 antiterminator hairpin, or Region 3:4 intrinsic terminator hairpin (followed by poly-U).
- High Tryptophan: Ribosome translates trpL rapidly, covering Region 2 as Region 3 and 4 are transcribed. Regions 3 and 4 pair to form the 3:4 Terminator Hairpin, terminating transcription prematurely before structural genes can be copied.
- Low Tryptophan: Ribosome stalls at the tandem Trp codons on Region 1 due to lack of Trp-tRNATrp. Stalled ribosome leaves Region 2 free to base-pair with Region 3, forming the 2:3 Antiterminator Hairpin. Formation of 2:3 precludes formation of the 3:4 terminator, permitting RNA Pol to transcribe the full operon.

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**[EXAMPLE 1: Deciphering the Wobble Hypothesis Pairing Rules]**

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Problem Context: A bacterial alanine tRNA has the anticodon sequence 5’-IGC-3’. Determine all the possible mRNA codons this single tRNA can recognize according to Crick’s Wobble Rules, and verify whether all recognized codons specify the amino acid Alanine.
Anticodon Analysis (5’-I1 - G2 - C3-3’):
- Position 3 of anticodon (5’ end is C3? No, reading 5’->3’: pos 1 is I, pos 2 is G, pos 3 is C).
- mRNA codon is antiparallel, reading 5’ to 3’:
- Codon Position 1 pairs with Anticodon Position 3 (C): standard C-G pairing -> Codon Pos 1 must be G.
- Codon Position 2 pairs with Anticodon Position 2 (G): standard G-C pairing -> Codon Pos 2 must be C.
- Codon Position 3 (the wobble position) pairs with Anticodon Position 1 (Inosine, I).
Wobble Base Pairing of Inosine (I):

- According to Crick's Wobble Hypothesis, Inosine at the 5' wobble position of an anticodon can form stable hydrogen bonds with three different bases at the 3' position of the codon: Uracil (U), Cytosine (C), and Adenine (A).
Resulting Recognized mRNA Codons:
1. 5'-G C U-3'
2. 5'-G C C-3'
3. 5'-G C A-3'
Codon Table Check for Alanine:
- Standard genetic code assigns GCU, GCC, GCA, and GCG to Alanine (a four-fold degenerate codon family).
Conclusion: This single tRNA(Ala) with 5'-IGC-3' anticodon successfully reads 3 out of the 4 Alanine codons without misreading any other amino acid, demonstrating how wobble minimizes the number of cellular tRNA species required.

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**[EXAMPLE 2: Diagnostic Phenotype Analysis of Lac Operon Merodiploid Mutants]**

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Problem Context: Using Jacob and Monod's classical merodiploid (F' plasmid) genetics, predict the beta-galactosidase (LacZ) and Permease (LacY) expression patterns (Constitutive, Inducible, or Uninducible) in the absence and presence of lactose for the following mutant strain: I+ O^c Z- Y+ / F' [ I- O+ Z+ Y- ].
Chromosome Genotype: I+ (functional repressor) | O^c (constitutive mutant operator, repressor cannot bind) | Z- (non-functional beta-gal) | Y+ (functional permease).
Episome (F') Genotype: I- (inactive repressor) | O+ (wild-type operator) | Z+ (functional beta-gal) | Y- (non-functional permease).
Regulatory Interaction Analysis:
1. Repressor (lacI): I+ is trans-dominant over I-. Functional repressor protein is produced from the chromosome, diffuses through the cytoplasm, and can bind to any accessible wild-type operator (O+).
2. Operator (lacO): Operators are cis-acting DNA elements; they only regulate structural genes physically contiguous on the same DNA strand.
- Evaluating Beta-Galactosidase (Z gene):
- Chromosomal Z is mutant (Z-).
- Functional Z+ is on the F' plasmid, controlled by wild-type operator O+.
- The diffusible I+ repressor binds F' O+, silencing Z+ in the absence of lactose.
- When lactose is added, allolactose induces repressor release, turning on Z+ synthesis.
- Therefore, Beta-Galactosidase is INDUCIBLE.
- Evaluating Lactose Permease (Y gene):
- F' plasmid Y is mutant (Y-).
- Functional Y+ is on the chromosome, directly attached downstream of the constitutive operator O^c.
- The I+ repressor cannot bind O^c. Transcription proceeds continuously whether lactose is present or absent.
- Therefore, Lactose Permease is CONSTITUTIVE.
Summary: Beta-Galactosidase = Inducible; Permease = Constitutive.

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**[EXAMPLE 3: Mechanism of Protein Synthesis Inhibitor Antibiotics]**

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Problem Context: Ribosomes are the primary target for over 50% of clinical antibiotics. Detail the molecular mechanisms of: (A) Chloramphenicol, (B) Tetracycline, (C) Streptomycin (Aminoglycoside), and (D) Puromycin.

Analysis of Antibiotic Targets:

1. Chloramphenicol: Binds reversibly to the 50S large ribosomal subunit in the Peptidyl Transferase Center cavity. It sterically blocks peptide bond formation by preventing the aminoacyl moiety of A-site aminoacyl-tRNA from interacting with the ester carbonyl of P-site peptidyl-tRNA.
2. Tetracycline: Binds reversibly to the 30S small ribosomal subunit near the A site. Physically blocks the entry of the ternary complex (EF-Tu-GTP-aa-tRNA), preventing aminoacyl-tRNA binding to the A site.
3. Streptomycin (Aminoglycoside): Binds to 16S rRNA within the 30S subunit decoding center. Induces conformational distortion that causes codon misreading (incorporating incorrect amino acids into growing chains) and prevents the transition from initiation complex to elongation at high concentrations.
4. Puromycin: Structural analog of the 3’ end of tyrosyl-tRNA. Enters the ribosomal A site and participates as an acceptor substrate in peptidyl transferase condensation. Forms a covalent peptidyl-puromycin bond; however, lacking an anticodon or CCA linkage, it triggers premature chain termination and releases truncated polypeptide fragments in both prokaryotes and eukaryotes.

Conclusion: Structural differences between 70S prokaryotic and 80S eukaryotic ribosomes provide the basis for selective antimicrobial therapy without host toxicity.

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**[EXAMPLE 4: Attenuation Modeling of the Tryptophan Operon]**

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Problem Context: In an engineered E. coli strain, the two tandem UGG codons in the trpL leader peptide are mutated to two Alanine codons (GCG-GCG). Predict the regulation of the trp operon structural genes under: (1) Starvation for Tryptophan, (2) Starvation for Alanine, and (3) Abundant nutrient conditions with all amino acids present.

Analysis of Leader Translation Dynamics:

Case 1 (Tryptophan Starvation, Alanine Abundant):

- Because the leader peptide now contains Ala codons instead of Trp codons, the ribosome does NOT stall at region 1 during tryptophan starvation.
- Charged Ala-tRNAAla is abundant; the ribosome translates rapidly through Region 1 and enters Region 2 before transcription of Region 3/4 is completed.
- The physical presence of the ribosome on Region 2 sterically prevents Region 2 from base-pairing with Region 3.
- Region 3 pairs with Region 4, forming the 3:4 Intrinsic Terminator Hairpin. Transcription is terminated.

- Outcome: The operon remains ATTENUATED (turned OFF) despite severe tryptophan starvation!
Case 2 (Alanine Starvation, Tryptophan Abundant):
- The ribosome arrives at the tandem GCG-GCG Alanine codons on Region 1 and stalls due to absence of charged Ala-tRNAAla.
- Stalling on Region 1 leaves Region 2 completely free and unshielded.
- As RNA Polymerase synthesizes Region 3, Region 2 and Region 3 spontaneously anneal to form the stable 2:3 Antiterminator Hairpin.
- Formation of 2:3 prevents the 3:4 terminator from forming. RNA Polymerase continues uninterrupted into the trpEDCBA structural genes.
- Outcome: Tryptophan biosynthetic enzymes are synthesized in response to Alanine starvation!
Case 3 (Abundant Nutrients):
- Ribosome translates leader rapidly without pause, covers Region 2, allowing 3:4 terminator hairpin formation.
- Outcome: Operon is attenuated (turned OFF).
Conclusion: This synthetic mutation proves that attenuation is a tRNA-sensing physical switch governed by ribosome stalling, independent of the downstream enzyme functions.

### >> 6.5 University Review & Examination Questions

* Q1: Detail the chemical mechanism of eukaryotic pre-mRNA splicing catalyzed by the spliceosome.
Answer: Eukaryotic splicing removes non-coding introns via two sequential transesterification reactions: (1) Assembly: U1 snRNP binds the 5’ splice site (GU), U2 snRNP binds the Branch Point Adenosine within the polypyrimidine tract. U4/U6-U5 tri-snRNP joins to assemble the pre-catalytic spliceosome (Complex B). U1 and U4 dissociate, activating catalytic Complex C. (2) First Transesterification: The nucleophilic 2’-OH group of the branch point adenosine attacks the 5’ phosphodiester bond at the 5’ splice site. This cleaves the exon 1-intron junction and forms an unusual 2’,5’-phosphodiester bond, creating a looped ’lariat’ intron intermediate and a free 5’ exon with a reactive 3’-OH group. (3) Second Transesterification: The liberated 3’-OH of exon 1 attacks the 3’ phosphodiester bond at the 3’ splice site. This ligates exon 1 to exon 2 via standard 3’,5’-phosphodiester linkage and releases the excised lariat intron, which is subsequently linearized by debranching enzymes and degraded.
* Q2: Explain the dual regulatory mechanisms controlling the E. coli Lactose Operon in the presence of various glucose and lactose concentrations.
Answer: The Lac operon is regulated by negative repressor control (LacI) and positive catabolite activator control (CAP-cAMP): (1) Glucose present, Lactose absent: LacI repressor binds operator lacO, preventing RNA Pol transcription. Low cAMP means CAP is inactive. Operon is completely repressed (Basal rate ~0). (2) Glucose present, Lactose present: Allolactose binds LacI, inducing derepression. However, because glucose is high, adenylate cyclase is inhibited, intracellular cAMP is low, and CAP cannot bind the promoter. RNA Pol binds poorly. Operon has very low, leaky expression. (3) Glucose absent, Lactose absent: LacI is bound to operator; even though cAMP is high and CAP-cAMP

is bound, transcription is blocked by the repressor. (4) Glucose absent, Lactose present: Derepression occurs (LacI released by allolactose) AND high cAMP drives CAP-cAMP binding to the promoter, bending DNA and activating RNA Pol. High-level maximal transcription of lacZYA ensues.

### >> 6.6 Key Takeaways and Revision Points

* Genetic code is triplet, non-overlapping, degenerate, and nearly universal; Wobble hypothesis allows 1 tRNA to read multiple codons.
* Ribosomal translation cycle consists of Initiation (Shine-Dalgarno, IFs), Elongation (EF-Tu, Peptidyl transferase ribozyme, EF-G), and Termination (RFs).
* Eukaryotic pre-mRNA processing involves 5’ m7G capping, 3’ polyadenylation (AAUAAA), and two-step spliceosomal lariat splicing.
* Jacob-Monod operon models: Lac operon is inducible (LacI derepression + CAP-cAMP catabolite activation); Trp operon uses corepressor TrpR and riboswitch-like leader peptide attenuation.